Recommendation System

Introduction

- Implicit（隱含式）Data：使用者活動如點擊的結連結、購買的產品　較容易蒐集
- Explicit（明確式）Data：評分、留言
- Collaborative Filtering：根據回應分類
- Content Based：根據劇情內容分類

Alogrithm　（尋找 implict data 的關聯性）

- Association Rule Mining（關聯規則探勘 ARM）：使用Apriori（先驗演算）
- Alternating Least Square（交替最小平方法 ALS）：collaborative filtering algorithm
- Content Based Filtering（內容過濾）：使用物品特色讓使用者被推薦到

Association Minig (Support and Confidence)

Support (支持度) $\text{Support}(A \to C) = P(A \cap C)$

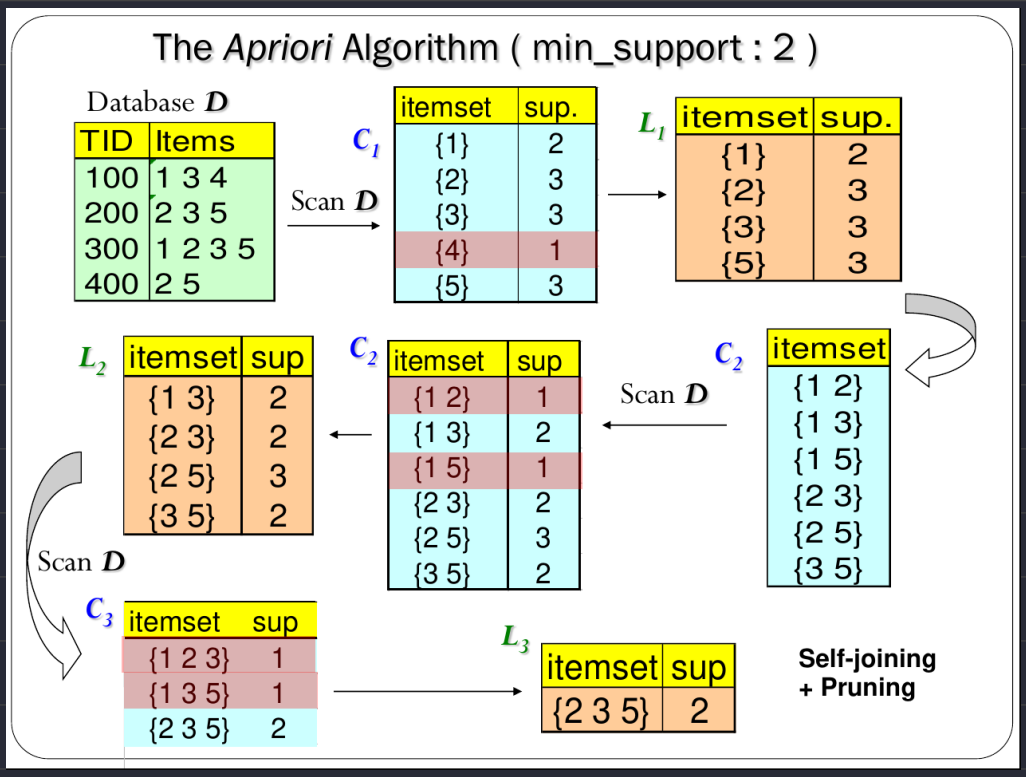
Confidence(信賴度) $\text{Confidence}(A \to C) = P(C|A) = P(A \cap C) \div P(A)$

Lift (增益率) : 規則多有效 $\text{Lift}(A \to C) = P(C|A) \div P(C) = P(A \cap C) \div [P(A) \times P(C)]$

Apriori algorithm

Transaction ID	Items Bought
2000	A,B,C
1000	A,C
4000	A,D
5000	B,E,F

Frequent Itemset	Support
{A}	75%
{B}	50%
{C}	50%
{A,C}	50%



Self-joining and Pruning:

$L_3 = \{abc, abd, acd, ace, bcd\}$

Self-Joining: $L_3 \cdot L_3 = \{abcd, abce, acde\}$

Pruning: abce is removed because abe is not in L3

acde is removed because ade is not in L3

Performance

Bad performance due to scanning database n+1 times (n is the longest pattern)

Frequent Patterns

Benifit

1. No candidate generation
2. Only need to scan database 2 times
3. Completeness (完整) and Compactness (緊密) in FP-Tree

Steps

1. Scan DB once, find frequent 1-itemset (single item pattern)
2. Order frequent items in frequency descending order
3. Scan DB again, construct FP-tree

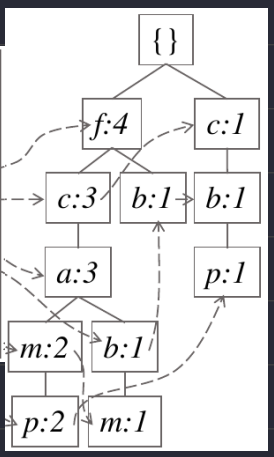
Example

Min-Support = 50% (count ≥ 3)

TID	Items bought	(ordered) frequent items
100	{f, a, c, d, g, i, m, p}	{f, c, a, m, p}
200	{a, b, c, f, l, m, o}	{f, c, a, b, m}
300	{b, f, h, j, o}	{f, b}
400	{b, c, k, s, p}	{c, b, p}
500	{a, f, c, e, l, p, m, n}	{f, c, a, m, p}

Header Table

Item	frequency	head
f	4	
c	4	
a	3	
b	3	
m	3	
p	3	

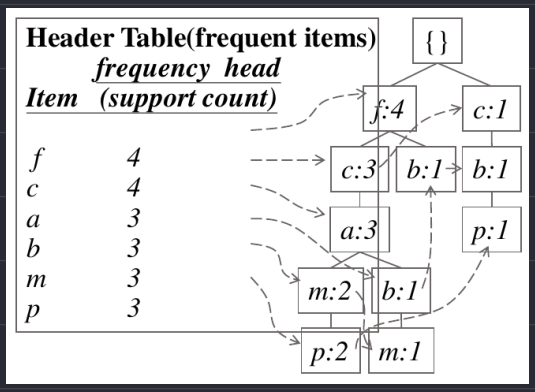


Mining Frequent Patterns Using FP-tree

1. Construct conditional pattern-base for each item
2. This will turn FP-Tree into conditional FP-Tree
3. Repeat the process on each newly created conditional FP-Tree
4. Stop until FP-tree is empty or it only contains 1 path

Example

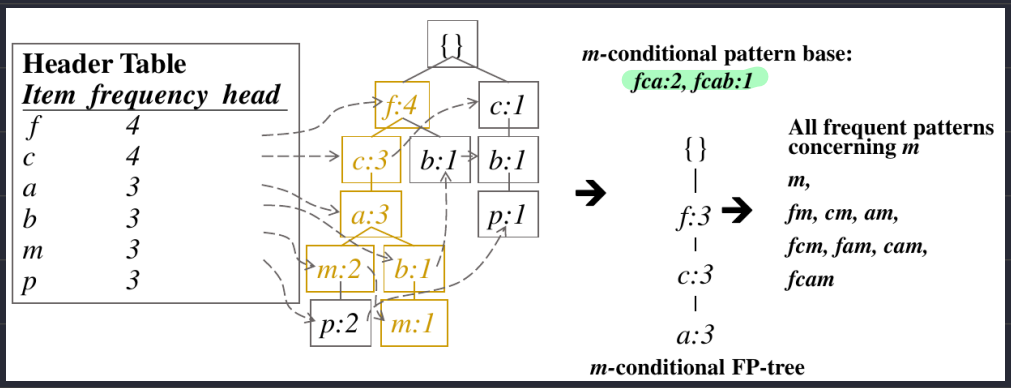
Step 1 From FP-tree to Conditional Pattern Base



Conditional pattern bases	
item	*cond. pattern base*
c	*f:3*
a	*fc:3*
b	*fca:1, f:1, c:1*
m	*fca:2, fcab:1*
p	*fcam:2, cb:1*

Step 2: Construct Conditional FP-tree for each children (min supp·=3)

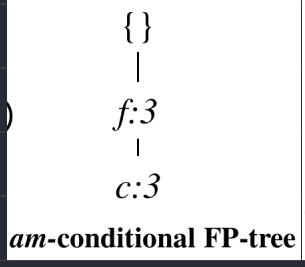
本張圖片僅展示 m-conditional base（從 m 向上）的 FP-Tree 建置方式

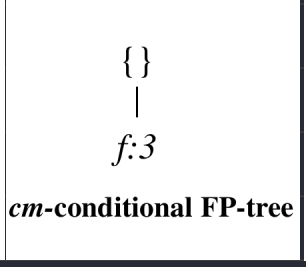


雖然 b 是 m 的母節點，但是因為其不符合 min-support 的要求，所以可以跳過

Item	Conditional pattern-base	Conditional FP-tree
p	{(fcam:2), (cb:1)}	{(c:3)}|p
m	{(fca:2), (fcab:1)}	{(f:3, c:3, a:3)}|m
b	{(fca:1), (f:1), (c:1)}	Empty
a	{(fc:3)}	{(f:3, c:3)}|a
c	{(f:3)}	{(f:3)}|c
f	Empty	Empty

Step 3 Recursively mine the conditional FP-tree





All frequent patterns concerning *m*

m,

fm, cm, am,

fcm, fam, cam,

fcam

Finding Similarity

User Collaborative Filtering

Jaccard Similarity

只觀察同樣的部分，完全不看兩者的數值大小關係

$J(A,B) = \frac{|A \cap B|}{|A \cup B|}$

Cosine Similarity

將雙方資料當成向量作內積計算

$\text{sim}(\vec{x}, \vec{y}) = \frac{\vec{x} \cdot \vec{y}}{\|\vec{x}\| \|\vec{y}\|} = \frac{\sum_{i=1}^{n} x_i y_i}{\sqrt{\sum_{i=1}^{n} x_i^2} \cdot \sqrt{\sum_{i=1}^{n} y_i^2}}$

Pearson Correlation Coefficient

$r_{xy} = \frac{\sum_{i \in I_{xy}} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i \in I_{xy}} (x_i - \bar{x})^2} \cdot \sqrt{\sum_{i \in I_{xy}} (y_i - \bar{y})^2}}$

Example

User	Item 1	Item 2	Item 3	Item 4	Item 5
u_1	⋆⋆⋆⋆⋆	⋆⋆⋆	–	⋆	⋆⋆⋆⋆
u_2	⋆⋆	–	⋆⋆⋆⋆⋆	⋆⋆⋆	⋆

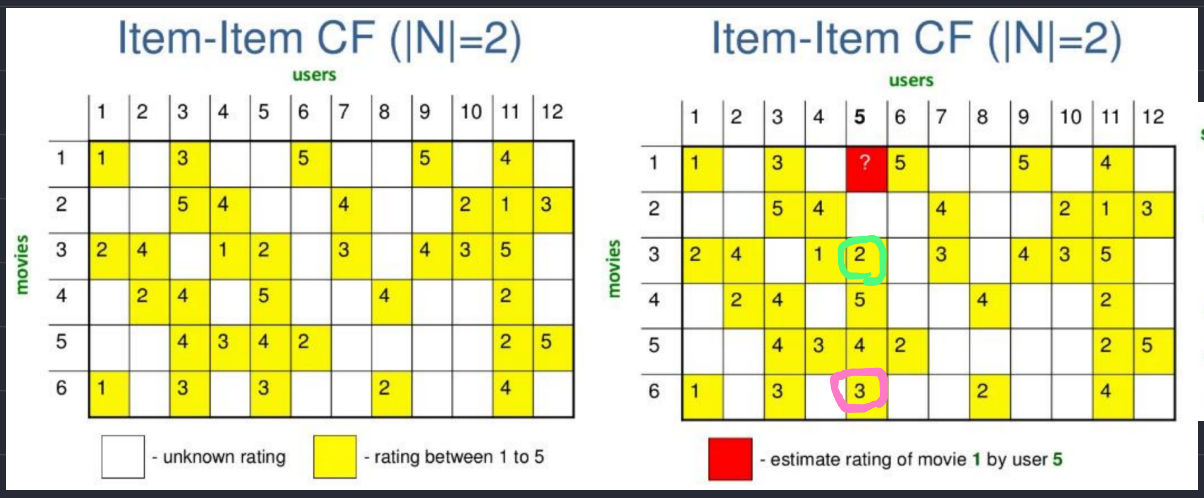
$r_1 = \{5, 3, 0, 1, 4\} \quad r_2 = \{2, 0, 5, 3, 1\}$

Jaccard: $J(u_1, u_2) = 3 \div 5$

Cosine: $\frac{r_1 \cdot r_2}{|r_1| \times |r_2|} = \frac{10 + 0 + 0 + 3 + 4}{\sqrt{51} \times \sqrt{39}} = \frac{17}{\sqrt{51} \times \sqrt{39}}$

Item-Based Collaborative Filtering

使用所有使用者給予的評價，比較物品之間的關聯性



Step 1: 找到與電影1相似的點影（假設是 3 與 6）

Step 2: 將 User 5 對 電影3 與 電影6 的評價讀取出來

Step 3: 帶入相似度進行加權平均 獲得 分數 $\hat{r}_{5,1} = \frac{(s_{1,3} \times r_{5,3}) + (s_{1,6} \times r_{5,6})}{s_{1,3} + s_{1,6}}$